



Bees can do math?

A study suggests that honeybees can grasp the concept of numerical symbols, and do basic math even. <https://digit.in/july19-47>



Recycling CO2 in the air

A new electrochemical process shortens the path to capturing and recycling CO2 from the air. <https://digit.in/july19-46>

come out with a process that completely eliminates the need for scaffolds in the process of 3D printing human tissues and organs.

AIM OF THE ALTERNATIVE SCAFFOLD-FREE METHOD

A group of scientists led by Professor Eben Alsberg, Richard and Loan Hill professor of Bioengineering and Orthopaedics at the University of Illinois, Chicago, aimed to engineer a scaffold-free method of building three-dimensional tissue which solely focused on building sophisticated structures (tissues and organs) that were functionally superior to the ones built using the traditional method. They believed that the advances made in 3D printing have brought remarkable capabilities to the practice of tissue engineering, however, a technology which can create and culture individual cell-based tissues without interfering scaffolds which hold the shape and architecture of the tissue has been unachievable until their innovation. With their alternative method that does away with scaffolds by using a hydrogel bead bath, they aim to achieve cell-to-cell communication which will let the tissue thrive and grow and eliminate toxic by-products by using their biodegradable solution.

THE PROCESS OF THE SCAFFOLD-FREE METHOD

The group of scientists led by Prof. Eben Alsberg developed an alternative method that allows for the production of 3D printed biological tissues without the need for scaffolds at all. These scientists have developed a system where they utilise a block of hydrogel which is made up of microscopic beads. In this block, a printing nozzle is lowered which moves back and forth - both vertically and horizontally - and deposits a 'bio-ink' which is entirely made up of stem cells. The bio-ink is firmly held in

place by the microbeads which ensure that the bio-ink stays exactly where it was deposited, with minimal resistance to the nozzle movement, within the three-dimensional chamber.



The bio-ink is deposited precisely within the hydrogel in this alternative method

Once the tissue cells are printed into the hydrogel bead solution, the whole matrix is then exposed to ultraviolet light. This, effectively, causes the beads to cross-link with each other which eventually causes them to freeze in place and hold shape. Now the printed cells are enabled to freely connect with one another, mature and grow within this newly-created stable structure. As they do so, technicians in the lab add in a nutrient-bath solution, which flows without restriction through the cross-linked beads and reaches the tissue. This solution provides fresh nutrients and allows them to dispose of waste products, which ultimately help them grow better.

Once the biotissue matures completely, the beads can be removed very easily either with gentle agitation or by allowing them to biodegrade harmlessly. The rate of this biodegradation can also be chemically controlled. Once the beads are removed or degrade naturally, the only thing left behind is the intact tissue or fully-formed organ. As of now, researchers have utilised this technique to successfully produce a rodent-sized femur as well as a cartilage ear.



Prof. Eben Alsberg led the team who created the scaffold-free method of 3D printing tissue

Alsberg said, "The hydrogel bead bath has unique properties which allow for both printing of the cell-only bio-ink in complex architectures,

and subsequent temporary stabilization of these cell-only structures to allow for cell-cell junctions to form." He states that the hydrogel bead bath used in his method is also temporary and doesn't create any toxic by-products, unlike traditional scaffolds supports. The stems used by Alsberg's team are stem cells which can differentiate into a wide variety of cells. The cells that were printed using this method, were reportedly, able to form stable, cell-to-cell connections through the specialised proteins.

IMPLICATIONS AND FUTURE APPLICATIONS

With this method, it has been demonstrated that cell aggregates can actually be managed and assembled to form larger functional tissues. It is a paradigm shift from how tissues and organs have been printed so far. While methods like creating 3D printed



World's first 3D printed heart with human cells and vessels

sugar scaffolds which dissolve after growing organs are quite viable in this field, this technique actually removes an entire process altogether - the creation of scaffolds - which makes it that much faster. 3D printing organs as per demand doesn't seem like a far-fetched idea now that tissue is actually being 3D printed successfully, whether it is with or without scaffolding. However, the removal of this process can ensure that the patients who need organs urgently can actually receive them that much faster. The method can also be extremely valuable for tissue engineering and regenerative medicine, drug screening and even as models to study developmental biology. 